

A Harnack answer to Mori's Problem 49

Later-time heat-kernel L^2 spaces are dense

Status. Literature-implied answer to the full problem. The decisive input is Li–Xu’s 2009 parabolic Harnack theorem; the implication below does not appear to be stated explicitly in the inspected literature.

Source and exact question

Shota Mori, *The heat kernel on $SL(2, \mathbb{R})$* , arXiv:1909.03670v3 [1], Section 8, Problem 49, PDF page 25:

Let $t, \varepsilon > 0$. Prove that the space

$$L^2\left(SL(2, \mathbb{R}), \rho_{SL(2, \mathbb{R})}(t + \varepsilon, g) dg\right)$$

is a dense subspace of

$$L^2\left(SL(2, \mathbb{R}), \rho_{SL(2, \mathbb{R})}(t, g) dg\right).$$

Identification

Mori’s metric is left invariant. Hence it is complete, and its Ricci tensor has a global lower bound $\text{Ric} \geq -kg$ for some $k \geq 0$: left translations are isometries, so the Ricci eigenvalues are the same at every point. Li and Xu’s parabolic Harnack inequality [2, Theorem 1.4] applies to every positive heat solution on such a complete noncompact manifold.

At the same spatial point $x_1 = x_2 = g$ and times $t_1 = t < t_2 = s$, their theorem gives

$$\rho_t(g) \leq C_{t,s} \rho_s(g) \quad (g \in G), \tag{1}$$

where, if $n = \dim G$,

$$C_{t,s} = \left(\frac{s}{t}\right)^{n/2} \left(\frac{1 + 2ks/3}{1 + 2kt/3}\right)^{-n/8} \exp\left(\frac{nk}{4}(s - t)\right) < \infty.$$

The distance term in the Harnack inequality vanishes, so the constant is uniform in g .

Theorem 1. *Let G be a connected noncompact Lie group with a left-invariant Riemannian metric, and let ρ_t be its positive heat kernel for $\partial_t u = \Delta u$. For $0 < t < s$, the natural inclusion*

$$L^2(G, \rho_s dg) \hookrightarrow L^2(G, \rho_t dg)$$

is continuous and has dense range. In particular, Mori’s Problem 49 has an affirmative answer by taking $G = SL(2, \mathbb{R})$ and $s = t + \varepsilon$.

Proof. The Harnack comparison (1) gives, for every $f \in L^2(G, \rho_s dg)$,

$$\|f\|_{L^2(\rho_t dg)}^2 = \int_G |f(g)|^2 \rho_t(g) dg \leq C_{t,s} \int_G |f(g)|^2 \rho_s(g) dg.$$

Thus the natural inclusion is well-defined and continuous. Both heat kernels are positive and continuous. Consequently $C_c(G) \subset L^2(G, \rho_s dg)$, while $C_c(G)$ is dense in $L^2(G, \rho_t dg)$ for the Radon measure $\rho_t(g) dg$. Hence the range of the inclusion is dense. $\square$

Provenance, scope, and verification

This is an agent-identified implication, not an original theorem claimed by this run. Li–Xu predates Mori’s question by ten years and therefore does not identify Problem 49. The proof resolves Problem 49 completely and in fact works on every connected noncompact Lie group with a left-invariant metric. It makes no claim about Mori’s Problems 46–48. For compact connected Lie groups, the same conclusion follows directly because the ratio of two positive continuous heat kernels is bounded on the compact group.

The hypothesis audit checks completeness, the Ricci lower bound, positivity, and matching heat-equation normalization. A bounded search of the run indexes, the exact wording of Problem 49, arXiv:1909.03670, and author/title variants found no paper explicitly stating this answer. The official source record is still v3 (2 October 2019). Human review should focus on the application of Li–Xu Theorem 1.4 to Mori’s normalization; both use $\partial_t u = \Delta u$ in the relevant statements.

References

- [1] S. Mori, *The heat kernel on $SL(2, \mathbb{R})$* , arXiv:1909.03670v3 (2019), Section 8, Problem 49.
- [2] J. Li and X. Xu, *Differential Harnack inequalities on Riemannian manifolds I: linear heat equation*, arXiv:0901.3849 (2009), Theorem 1.4.